Multi-Scale Regime Detection with Fuzzy Aggregation for Adaptive Sector Portfolio Management

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Abstract

We propose a multi-scale regime detection framework that fuses four complementary detectors—CUSUM sequential analysis on z-score returns, EWMA volatility crossover, Hamilton Markov-switching with a recursive out-of-sample filter, and PELT structural break detection—into a unified stress probability via Takagi-Sugeno fuzzy inference with bounded L-BFGS-B optimization. The composite regime signal drives an adaptive sector portfolio strategy that classifies eleven S&P 500 sector ETFs into three baskets (Tactical, Avoid, Core) based on GARCH(1,1)-t conditional volatility terciles and Ornstein-Uhlenbeck recovery half-life, with implicit cash allocation as the primary risk-reduction mechanism. In a walk-forward backtest spanning 2009–2025 with quarterly recalibration, the framework achieves a Sharpe ratio of 0.78, a maximum drawdown of -19.60% (versus -41.71% for SPY), and a Calmar ratio of 0.59, while capturing 98% of the benchmark return at 73% of its volatility. All thresholds and parameters are estimated from data using documented statistical methods, with zero lookahead bias enforced throughout.

Keywords: regime detection, fuzzy inference, portfolio management, CUSUM, Markov switching, GARCH, walk-forward validation, implicit cash allocation

1 Introduction

Financial markets exhibit regime-dependent behavior characterized by distinct periods of calm, transition, and stress [Hamilton, 1989]. Traditional portfolio strategies that assume stationary return dynamics suffer significant drawdowns during regime shifts, as the 2008 Global Financial Crisis, the 2020 COVID-19 crash, and the 2022 interest rate hiking cycle demonstrated. Even diversified portfolios that hold all sectors with equal or risk-proportional weights cannot avoid the synchronous drawdowns that characterize systemic stress events.

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The central challenge in regime-adaptive portfolio management is *timely* and *reliable* regime detection. Individual detectors operate on different time-scales and suffer from distinct failure modes: short-term indicators produce false alarms during idiosyncratic volatility, while long-term models react too slowly to protect against rapid drawdowns. The secondary challenge is translating regime signals into portfolio actions that actually reduce risk: a system that detects stress but merely redistributes weight among risky assets (rather than moving to cash) provides limited drawdown protection.

This paper addresses both challenges. We propose an ensemble architecture that fuses four complementary detectors through a Takagi-Sugeno fuzzy inference system [Takagi and Sugeno, 1985], combined with an adaptive basket allocation framework where implicit cash allocation serves as the primary risk-reduction mechanism.

Our contributions are fourfold:

1. A multi-scale detector ensemble spanning ultra-short (5-day) to long-term (252+ day) horizons, with CUSUM operating on z-score-standardized returns for scale invariance and Markov-switching using a recursive Hamilton filter for real-time out-of-sample updates.

2. A fuzzy aggregation framework that learns optimal detector weights and membership functions by minimizing the Brier score against realized drawdowns, with bounded optimization and minimum weight floors to prevent detector suppression.

3. A tercile-based basket construction methodology that classifies sector ETFs using GARCH(1,1)-t conditional volatility and Ornstein-Uhlenbeck recovery dynamics, with graduated Core-basket response to extreme stress.

4. An implicit cash mechanism where stress-induced weight reduction produces sub-unity portfolio allocation, with the deficit earning the risk-free rate rather than being redistributed to other risky assets.

2 Related Work

2.1 Regime Detection

Hamilton [1989] introduced Markov-switching models for business cycle analysis, establishing the foundation for regime-based asset allocation. Ang and Bekaert [2002] extended this framework to international asset allocation, demonstrating that regime-switching models can improve out-of-sample portfolio performance. A persistent challenge is that single-scale detectors miss signals outside their native frequency range, motivating multi-scale ensemble approaches.

2.2 Fuzzy Systems in Finance

Gradojevic and Gençay [2013] demonstrated the effectiveness of fuzzy logic systems for financial decision-making under uncertainty. The Takagi-Sugeno framework [Takagi and Sugeno, 1985] pro-

vides a principled approach to combining multiple information sources with calibratable membership functions. Our work extends this by jointly optimizing membership parameters and detector weights with explicit bounds to prevent degenerate solutions.

2.3 Adaptive Portfolio Management

Nystrup et al. [2017] showed that regime-based asset allocation outperforms static strategies when the regime model is properly calibrated. Kirby and Ostdiek [2012] demonstrated that simple timing strategies based on inverse-volatility weighting outperform naïve diversification. Our framework builds on inverse-volatility weighting as the base allocation and adds regime-dependent scaling with an implicit cash buffer.

2.4 Change-Point Detection

Page [1954] introduced the CUSUM procedure for sequential change detection. Killick et al. [2012] proposed the PELT algorithm for optimal change-point detection with linear computational cost. Siegmund [1985] derived ARL approximations for setting CUSUM decision thresholds. We apply CUSUM to z-score-standardized returns to achieve scale-invariant detection thresholds.

3 Methodology

3.1 Multi-Scale Detector Ensemble

We deploy four detectors targeting complementary time-scales. All parameters are estimated from training data at each walk-forward step.

3.1.1 CUSUM Sequential Detector (5–10 day)

Following Page [1954], we maintain upward and downward CUSUM accumulators on *z-score-standardized* returns:

$$z_t = \frac{r_t - \hat{\mu}_0}{\hat{\sigma}_{\text{train}}} \tag{1}$$

$$S_t^+ = \max(0,\ S_{t-1}^+ + z_t - k) \tag{2}$$

$$S_t^- = \max(0,\ S_{t-1}^- - z_t - k) \tag{3}$$

where $\hat{\mu}_0$ and $\hat{\sigma}_{\text{train}}$ are the mean and standard deviation of training-window returns, $k = 0.5$ (detecting a 1σ shift in standardized space), and the decision threshold is $h = (\ln(\text{ARL}_0) + 1.166)/\delta$ with $\delta = 1$ per Siegmund [1985], yielding $h \approx 6.70$ for a target average run length of 252 trading days. The signal is $\min(\max(S_t^+, S_t^-)/h,\ 1)$.

Operating in z-score space makes the threshold scale-invariant: a single -5% daily return (approximately $z = -4.2$) produces a signal of ~ 0.55, while three consecutive -2% days ($z \approx -1.7$

each) yield a signal of ~ 0.54. Under the original raw-return formulation, the threshold was approximately 558, requiring over 12,000 consecutive crash days to trigger—rendering the detector inoperative.

3.1.2 EWMA Volatility Crossover (21–42 day)

Two EWMA volatility estimators with decay factors λ_{fast} and λ_{slow} are calibrated via maximum likelihood estimation [Longerstaey and Spencer, 1996]:

$$\hat{\sigma}^2_{\cdot,t} = \lambda_{\cdot}\, \hat{\sigma}^2_{\cdot,t-1} + (1 - \lambda_{\cdot})\, r_t^2 \tag{4}$$

with $\lambda_{\text{fast}} \in [0.85, 0.96]$ and $\lambda_{\text{slow}} \in [0.95, 0.995]$. The crossover ratio $(\hat{\sigma}_{\text{fast}} - \hat{\sigma}_{\text{slow}})/\hat{\sigma}_{\text{slow}}$ is normalized to $[0, 1]$ via the empirical CDF of training-window values.

3.1.3 Hamilton Markov-Switching with Recursive Filter (63–126 day)

We fit a 2-state Markov-switching regression [Hamilton, 1989] on weekly SPY returns:

$$r_t \mid S_t = i \;\sim\; \mathcal{N}(\mu_i, \sigma_i^2), \quad i \in \{0, 1\} \tag{5}$$

The model is estimated on the training window via EM. During the out-of-sample window, rather than holding the filtered probability constant (which would eliminate the detector's ability to react to new information), we implement a *recursive Hamilton filter update*. Daily returns are accumulated into a pseudo-weekly buffer, and every 5 trading days the filter is updated:

1. **Prediction:** $\boldsymbol{\xi}_{t|t-1} = \mathbf{P}^\top \boldsymbol{\xi}_{t-1|t-1}$ where $\mathbf{P}$ is the fitted transition matrix.
2. **Likelihood:** $\eta_j = \phi(r_t^{(w)}; \mu_j, \sigma_j^2)$ for each state j, where $r_t^{(w)}$ is the accumulated pseudo-weekly return.
3. **Update:** $\boldsymbol{\xi}_{t|t} = (\boldsymbol{\xi}_{t|t-1} \odot \boldsymbol{\eta})/(\mathbf{1}^\top(\boldsymbol{\xi}_{t|t-1} \odot \boldsymbol{\eta}))$.

The signal is $P(S_t = \text{stress} \mid r_1, \ldots, r_t)$, using only contemporaneous and past data. Smoothed probabilities, which condition on the full sample, are not used to prevent lookahead bias.

3.1.4 Structural Break Detection (252+ day)

Using the PELT algorithm [Killick et al., 2012], we detect structural breaks in the cumulative return series with a BIC-calibrated penalty $\lambda = \ln(n) \cdot \hat{\sigma}^2$. The signal decays exponentially from the most recent break: $s_t = \exp(-d_t/\tau)$ where d_t is the number of days since the last break and $\tau = 21$.

3.2 Fuzzy Aggregation Framework

Each detector signal $x_i \in [0, 1]$ passes through a calibrated sigmoid membership function [Takagi and Sugeno, 1985]:

$$\mu_{\text{stress}}(x_i) = \frac{1}{1 + \exp(-a_i(x_i - c_i))} \tag{6}$$

where a_i (steepness) and c_i (crossover point) are per-detector parameters. The composite stress probability is:

$$P(\text{stress}) = \sum_{i=1}^{4} w_i \cdot \mu_{\text{stress}}(x_i), \quad \sum w_i = 1,\ w_i \geq w_{\min} \tag{7}$$

Parameters $\{a_i, c_i, w_i\}$ are jointly optimized by minimizing the Brier score [Brier, 1950] against realized drawdowns in the training window:

$$\mathcal{L}_{\text{Brier}} = \frac{1}{T} \sum_{t=1}^{T} \left(P_{\text{composite},t} - D_t\right)^2 \tag{8}$$

where $D_t = 1$ if a drawdown exceeding the 10th percentile of training-window drawdowns occurs within the next 21 trading days.

Optimization uses L-BFGS-B with explicit bounds: $a_i \in [1, 50]$, $c_i \in [0.05, 0.95]$, and raw weights $\tilde{w}_i \in [-5, 5]$ (mapped through softmax). A minimum weight floor of $w_{\min} = 0.05$ is applied post-softmax:

$$w_i = w_{\min} + (1 - 4w_{\min}) \cdot \text{softmax}(\tilde{w})_i \tag{9}$$

This ensures no detector is suppressed below 5% weight regardless of the optimization outcome, preventing degenerate solutions where the optimizer concentrates on a single detector.

3.3 Asset Characterization

3.3.1 GARCH(1,1)-t Conditional Volatility

For each sector ETF, we estimate [Bollerslev, 1986, Engle, 1982]:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \quad z_t \sim t_\nu \tag{10}$$

yielding conditional VaR(99%) $= \hat{\mu} + \hat{\sigma}_t \cdot t_\nu^{-1}(0.01) \cdot \sqrt{(\nu - 2)/\nu}$.

3.3.2 Ornstein-Uhlenbeck Recovery Half-Life

We model drawdown recovery as a discretized OU process [Uhlenbeck and Ornstein, 1930]:

$$\Delta X_t = \kappa(\theta - X_{t-1})\Delta t + \varepsilon_t \tag{11}$$

The half-life $\ln(2)/\kappa$ measures recovery speed. Sectors with $\kappa \leq 0$ are flagged as non-mean-reverting.

3.4 Adaptive Basket Construction

At each rebalance date, sectors are classified into three baskets using data-driven boundaries. We use *tercile* boundaries on conditional volatility (33rd and 67th percentiles of the cross-sectional vol distribution) combined with median half-life:

- **Basket A (Tactical):** Top or middle tercile vol with fast recovery (HL $<$ median). Liquidated when $P(\text{stress})$ exceeds the calibrated entry threshold; re-entered when $P(\text{stress})$ falls below the exit threshold.
- **Basket B (Avoid):** Top tercile vol with slow recovery (HL $\geq$ median). Continuously de-risked: $w_i = w_i^{\text{base}} \cdot (1 - P(\text{stress}))$.
- **Basket C (Core):** Bottom tercile vol. Held at full weight through mild stress. Above $P(\text{stress}) = 0.7$, graduated linear scaling:

$$w_i = w_i^{\text{base}} \cdot \max\left(0,\ \frac{1 - P(\text{stress})}{0.3}\right) \tag{12}$$

  This preserves stability during moderate corrections while enabling full de-risking during tail events.

Base weights are computed using inverse-volatility weighting [Kirby and Ostdiek, 2012] *normalized to sum to 1.0 before any stress scaling is applied.* This normalization is critical: it ensures that the stress-scaled weights have an interpretable sum less than or equal to 1.0, where the deficit represents implicit cash.

3.5 Implicit Cash Allocation

The portfolio's risk-reduction mechanism operates through implicit cash. After basket-specific stress scaling, the sum of portfolio weights is:

$$W_{\text{total}} = \sum_i w_i \leq 1.0 \tag{13}$$

The deficit $1 - W_{\text{total}}$ earns the daily risk-free rate. This design avoids a critical pitfall of weight-normalization approaches: when stressed assets are de-weighted and the result is renormalized to sum to 1.0, the freed capacity is *redistributed* to remaining assets, potentially increasing concentration risk during the exact periods when diversification matters most. By allowing sub-unity exposure, the system genuinely reduces market risk during stress.

3.6 Walk-Forward Validation

The entire system is validated using expanding-window walk-forward analysis with a 504-day (2-year) minimum training window and 63-day (quarterly) out-of-sample steps. At each step, *all*

parameters are re-estimated from training data only: detector calibration, fuzzy membership optimization, GARCH fitting, half-life estimation, basket classification, and entry/exit threshold calibration. The threshold calibration uses a grid search over the training-window Sharpe ratio. Transaction costs of 10 basis points per trade are applied to all weight changes.

4 Results

4.1 Data

We use daily close prices for eleven S&P 500 sector ETFs (XLK, XLF, XLE, XLV, XLI, XLY, XLP, XLU, XLRE, XLB, XLC) plus SPY as benchmark, spanning January 2007 to December 2025. Fama-French five-factor data [Fama and French, 2015] provides the daily risk-free rate.

4.2 Portfolio Performance

Table 1: Walk-Forward Out-of-Sample Performance (2009–2025)

Metric	**Regime-Adaptive**	**SPY B&H**	**Equal-Weight**
Ann. Return	11.61%	11.79%	8.52%
Ann. Volatility	13.22%	17.98%	16.05%
Sharpe Ratio	**0.78**	0.58	0.45
Sortino Ratio	**0.75**	0.54	0.42
Calmar Ratio	**0.59**	0.28	0.18
Max Drawdown	**−19.60%**	−41.71%	−46.40%
Max DD Duration	**226d**	512d	530d
Ann. Turnover	4.01	—	—
Regime Transitions	116	—	—
Hit Rate	91.6%	—	—
False Alarm Rate	8.4%	—	—

Table 1 summarizes the out-of-sample results. The regime-adaptive strategy captures 98.5% of SPY's annualized return (11.61% vs. 11.79%) while reducing volatility by 26% (13.22% vs. 17.98%) and maximum drawdown by 53% (−19.60% vs. −41.71%). The Sharpe ratio of 0.78 exceeds SPY's 0.58 by 34%, driven entirely by volatility reduction rather than return enhancement. The Calmar ratio more than doubles that of the benchmark (0.59 vs. 0.28), indicating substantially improved drawdown-adjusted performance.

The strategy achieves a 91.6% hit rate on stress detection with an 8.4% false alarm rate. The modest increase in false alarms relative to a purely threshold-based system is an acceptable cost for the improved reactivity of the z-score CUSUM and recursive Markov filter.

4.3 Detector Contribution Analysis

The Brier-score-optimized weights with minimum floor reveal the relative importance of each time-scale. CUSUM receives the dominant weight ($\sim$78%), reflecting the value of rapid detection in the z-score formulation. The structural break detector contributes $\sim$12%, capturing long-horizon regime changes. EWMA and Markov-switching each contribute at or near the 5% floor, providing supplementary information that improves the composite signal's reliability without dominating it.

The shift from EWMA dominance (in the pre-fix system) to CUSUM dominance is a direct consequence of the z-score standardization: the CUSUM detector now produces a signal with meaningful dynamic range (0 to 1.0 within days of a crash), whereas previously its maximum signal was 0.001.

4.4 Cash Allocation Dynamics

The implicit cash mechanism is the primary driver of drawdown reduction. Figure **??** illustrates the portfolio's total equity exposure during major stress events.

During the COVID-19 crash (February–March 2020), the portfolio's equity exposure dropped to 21.1% (78.9% implicit cash) within approximately one week of the drawdown onset. The CUSUM detector fired first (signal ≈ 0.55 on the initial -3.4% day on February 24), followed by the recursive Markov filter reaching 0.98 stress probability within 5 trading days. Recovery re-entry occurred gradually through April–May 2020 as P(stress) declined.

During the 2022 rate-hiking drawdown, the structural break detector complemented the CUSUM signal by identifying a regime change in cumulative returns, maintaining elevated stress probability through the extended drawdown period.

4.5 Basket Classification Stability

The tercile-based classification produces a typical distribution of 3–4 assets in Basket A (Tactical), 1–3 in Basket B (Avoid), and 4–6 in Basket C (Core), providing the strategy with 5–7 actively managed positions at each rebalance. Energy (XLE) and Financials (XLF) frequently appear in Baskets A and B respectively, consistent with their high conditional volatility and distinct recovery characteristics.

5 Discussion

5.1 The Normalization Problem

A key design insight is that portfolio weight normalization can *negate* regime-responsive behavior. In the standard approach, inverse-volatility weights for N assets produce raw sums of $\sim$ 20–30. When stressed assets are de-weighted and the result is normalized to sum to 1.0, the freed capacity is redistributed proportionally to all surviving positions—including those in the "hold" basket. This means the portfolio *increases* its allocation to Core assets during stress, which is the opposite of risk

reduction. Our approach normalizes base weights to 1.0 before stress scaling, so that post-scaling sums below 1.0 represent genuine cash allocation.

5.2 Detector Scale Invariance

The z-score standardization of CUSUM inputs is critical for practical deployment. With raw returns, the Siegmund threshold scales inversely with the standard deviation of daily returns ($h \propto 1/\sigma$), producing thresholds of $\sim$558 for typical equity index returns ($\sigma \approx 0.012$). Daily accumulator increments of $\sim$0.04 require thousands of consecutive crash days to trigger. In z-score space, the threshold is $\sim$6.70 regardless of the return scale, and a single -5% day ($z \approx -4.2$) produces a meaningful signal of 0.55.

5.3 Limitations

The framework has several limitations. First, the annualized turnover of $4.01\times$ creates transaction cost drag. While the 10 bps assumption is conservative for liquid ETFs (typical bid-ask spreads of 1–2 bps), higher-frequency strategies or less liquid instruments would face larger friction. Second, the quarterly recalibration frequency may lag rapidly evolving market dynamics; event-driven recalibration could improve responsiveness. Third, the CUSUM detector's dominance in the post-fix ensemble suggests that the remaining three detectors may contribute primarily through ensemble diversity rather than independent predictive power.

5.4 Robustness

All parameters are re-estimated at each walk-forward step using only past data, ensuring that the results are not affected by lookahead bias. The use of multiple independent detectors provides robustness to individual detector failure modes. The bounded L-BFGS-B optimization with minimum weight floors prevents the optimizer from converging to degenerate single-detector solutions. The implicit cash mechanism provides a natural risk-reduction pathway that does not depend on correctly predicting the *type* of stress, only its *presence*.

6 Conclusion

We have presented a multi-scale regime detection framework that fuses four complementary detectors through Takagi-Sugeno fuzzy inference for adaptive sector portfolio management. The walk-forward validated results demonstrate that the framework achieves a Sharpe ratio of 0.78 with a maximum drawdown of -19.60%—a 53% reduction relative to the SPY benchmark—while maintaining nearly identical annualized returns. The key innovations are: (1) z-score standardization of CUSUM inputs for scale-invariant detection, (2) recursive Hamilton filtering for real-time Markov-switching updates, (3) bounded fuzzy optimization with minimum detector weight floors, and (4) implicit cash allocation through sub-unity weight sums rather than forced normalization.

The framework's success lies not in predicting market direction but in reliably detecting the *onset* of stress and responding with genuine risk reduction. During the COVID-19 crash, the portfolio moved to 79% cash within days of the drawdown onset and gradually re-entered during recovery. This asymmetric response—fast to de-risk, gradual to re-risk—is the primary driver of the improved risk-adjusted performance.

Future work may explore: (1) event-driven recalibration triggered by detector alerts rather than fixed quarterly steps; (2) cross-sectional regime signals from equity factor models; (3) extension to multi-asset-class allocation including fixed income and commodities; and (4) incorporation of options-based tail hedging as a complement to the implicit cash mechanism.

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